
Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

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Abstract

Conformal prediction provides distribution-free coverage guarantees, but these degrade under distribution shift—and practitioners lack tools to anticipate which deployed models will fail before observing test data. We propose SHapley Additive exPlanations (SHAP) concentration—the fraction of feature importance concentrated in the top feature—as a pre-deployment diagnostic for conformal prediction vulnerability in gradient-boosted classifiers. Using COVID-19 as a naturalistic case study, eight supply chain tasks experience identical temporal shift yet coverage drops ranging from negligible to catastrophic. Feature-importance concentration is strongly associated with failure severity across 16 multiclass tasks in 9 domains, while standard distributional shift detectors detect shift uniformly across tasks but cannot distinguish catastrophic from robust outcomes. External validation across 9 non-supply-chain datasets shows partial transfer. We prove a formal theorem showing that Adaptive Prediction Sets conformity-score bounds worsen monotonically with concentration under explicit assumptions, verified empirically. A decision framework operationalizes the diagnostic as an exploratory pre-deployment rule with an uncertainty band around a concentration threshold.

1 INTRODUCTION

Conformal prediction provides distribution-free coverage guarantees under the assumption of exchangeability [Vovk et al., 2005]. However, real-world deployments face distribution shifts that violate this assumption. While prior work characterizes *how* conformal prediction degrades under shift [Tibshirani et al., 2019, Barber et al., 2023], a critical gap remains: **Can we identify which deployed models**

will fail before observing test data?

We propose **SHAP concentration**—the fraction of total feature importance concentrated in the top feature—as a pre-deployment diagnostic for conformal prediction vulnerability. Using a supply chain benchmark with temporal splits aligned to COVID-19 (February–July 2020), we conduct an observational case study where 8 classification tasks experience identical temporal shift yet exhibit coverage drops ranging from negligible ($<1\%$) to catastrophic ($>70\%$).

Our contributions:

1. **Pre-deployment diagnostic:** SHAP concentration—the fraction of importance in the top feature—correlates with conformal failure severity: $\rho = 0.853$, $p < 0.001$ across 16 multiclass tasks in 9 domains (Kendall $\tau = 0.667$; bootstrap 95% CI $[0.50, 0.96]$); $\rho = 0.833$, $p = 0.010$ within supply chain ($n = 8$, stable under LOO: $\rho \in [0.75, 0.96]$). The 40% threshold transfers to 9 non-supply-chain datasets without tuning (7/9 deterministic, Covertype correctly flagged at $C = 49.8\%$). A model-sensitivity analysis ($\rho = 0.833$ LightGBM vs. $\rho = 0.30$ RF, $\rho = 0.43$ MLP) identifies the scope: the diagnostic is specific to gradient-boosted classifiers where sequential training produces genuine single-feature dependence that bagging (RF) dilutes and MLP-SHAP approximations obscure (Section 3.6).
2. **Formal theory and comparative evidence:** Theorem 1 establishes that APS score and coverage bounds worsen monotonically with C (Section 4), verified on all 5 applicable tasks; catastrophic tasks show *decreasing* prediction entropy (confident misclassification). Standard shift indicators (MMD, C2ST, PSI¹) detect shift for all 8 tasks uniformly yet correlate $\rho \leq 0.19$ with coverage drop—shift detection does not predict failure severity.

¹Population Stability Index (PSI): a monitoring metric standard in credit scoring practice [Siddiqi, 2006]. $\text{PSI} > 0.002$ indicates detectable nonzero shift; conventional significance threshold is > 0.1 .

3. **Operational framework (exploratory):** A three-step protocol (compute C , check protective factors, monitor) with an uncertainty band around the 40% threshold. Retraining partially recovers coverage (+19 pp, $p = 0.036$ unadjusted, single seed; Holm-corrected $p = 0.11$); mitigation is mechanism-dependent (Appendix E).

2 RELATED WORK

Conformal Prediction. Vovk et al. [2005] introduced conformal prediction with exchangeability guarantees; Shafer and Vovk [2008] provide an accessible tutorial. Recent work extends to classification [Romano et al., 2020, Cauchois et al., 2021] and regression [Romano et al., 2019, Lei et al., 2018]. Angelopoulos and Bates [2023] provide a comprehensive survey. Ding et al. [2023] show that class-conditional coverage guarantees become harder to achieve as class count grows and propose clustered conformal prediction as a remedy; Gibbs et al. [2025] provide finite-sample conditional guarantees via a covariate-shift reweighting framework interpolating between marginal and conditional validity. Both works frame class count as a challenge—a confound we explicitly examine via partial correlation analysis (Section 5.4).

Conformal Prediction Under Shift. Tibshirani et al. [2019] study covariate shift with known propensity scores. Podkopaev and Ramdas [2021] develop distribution-free uncertainty quantification for classification under label shift. Barber et al. [2023] provide a comprehensive treatment beyond exchangeability, bounding coverage loss by the total variation distance between test and calibration score distributions. Kasa and Taylor [2023] empirically characterize how CP degrades across many vision architectures and shift types, showing substantial degradation is common—our work addresses the complementary question of which tabular models will fail *before* test data is observed. We complement these theoretical characterizations with an empirical diagnostic that is associated with this distributional divergence—and hence with which tasks fail—before deployment.

Adaptive Methods. Gibbs and Candès [2021, 2024] develop Adaptive Conformal Inference (ACI) for non-stationary settings, with extensions by Zaffran et al. [2022] for time series, Feldman et al. [2023] for online risk control, and Bhatnagar et al. [2023] for strongly adaptive online learning. Angelopoulos et al. [2024] generalize conformal prediction to broader risk measures. Kasa et al. [2025] propose entropy-based methods (ECP/EACP) that adapt prediction sets using unlabeled test data at deployment time—a related but distinct goal from ours, which requires no test observations: SHAP concentration is a structural pre-deployment diagnostic computed solely on validation data. Our experiments show ACI recovers marginal coverage un-

der severe shift but at a substantial cost in prediction set informativeness.

Shift Detection. WILDS [Koh et al., 2021] and Shifts [Malinin et al., 2021] provide benchmarks, while Gulrajani and Lopez-Paz [2021] show domain generalization methods often fail to improve over ERM under realistic distribution shifts. Gardner et al. [2023] benchmark tabular distribution shift across 15 tasks—the closest tabular-specific benchmark; our SALT dataset differs in providing a naturalistic COVID-19 temporal split with documented business-disruption context rather than curated benchmark partitions. Garg et al. [2022] use unlabeled test data to predict accuracy degradation—a related goal but requiring test-time observations. Kernel-based tests (MMD [Gretton et al., 2012]) and classifier two-sample tests (C2ST [Lopez-Paz and Oquab, 2017]) detect distributional differences but do not predict the *severity* of downstream failure. We show empirically that MMD and C2ST detect shift for all tasks uniformly yet cannot distinguish catastrophic from robust conformal outcomes (Section 6.3). Our contribution is a *pre-deployment* diagnostic using feature importance structure that predicts failure severity, not merely shift presence. Miller et al. [2021] show that in-distribution (ID) validation accuracy predicts out-of-distribution (OOD) accuracy for image classifiers—a correlation that does not hold for conformal coverage in our tabular setting, where ID validation coverages cluster near the 90% target, spanning 83.6%–99.9% (Table 1), yet test-time coverage spans 12–100%, reinforcing the need for structure-based diagnostics beyond accuracy metrics.

Interpretability for Reliability. Our work connects SHAP [Lundberg and Lee, 2017, Lundberg et al., 2020] to model reliability assessment, extending feature attribution from post-hoc analysis to prospective failure diagnosis. Our use is prospective: concentration is computed pre-deployment on validation data, predicting which models will fail.

3 METHODOLOGY

3.1 DATASET AND TEMPORAL SPLITS

We use the SALT (Supply chain ALlocATion) dataset from RelBench [Robinson et al., 2024], a supply chain benchmark with temporal splits. All 8 classification tasks from the rel-salt benchmark are included; no tasks were excluded. External validation uses 9 additional datasets spanning 9 non-supply-chain benchmarks. The primary endpoint includes only $K \geq 4$ class tasks: near-binary tasks ($K \leq 3$) are excluded because binary APS prediction sets ($\{0\}$, $\{1\}$, $\{0, 1\}$) have a structural ceiling that blocks the concentration mechanism (Contribution 1); this criterion follows from APS mechanics, not from the data. Stack Overflow ($K = 3$) falls under this exclusion. Applying it yields 8 external mul-

ticlass datasets; together with the 8 multiclass SALT tasks this gives the primary endpoint of $n = 16$ multiclass tasks across 9 domains (SALT counts as one domain).

- **Train:** Before February 2020 (pre-COVID)
- **Validation:** February–July 2020 (COVID onset)
- **Test:** After July 2020 (COVID peak)

Data separation protocol. SHAP concentration and the 40% threshold are computed exclusively on validation-time data. The test set is used only for final evaluation and never informs diagnostic thresholds. For threshold evaluation we use at-risk labels defined by observed test coverage drops (drop > 15 pp); the threshold sensitivity analysis in Appendix D is therefore post-hoc validation, not prospective assessment.

3.2 CONFORMAL PREDICTION SETUP

We use Adaptive Prediction Sets (APS) [Romano et al., 2020] with $\alpha = 0.1$ (90% target coverage). We use the non-randomized variant (without the uniform tie-breaking variable U), which provides conservative rather than exact coverage. This choice does not affect the diagnostic analysis, as the same scoring rule is applied to both calibration and test sets.

1. Train LightGBM classifier with fixed hyperparameters
2. Calibrate conformal predictor on 50% of validation set
3. Evaluate coverage on held-out validation and test sets

3.3 ENSEMBLE PROTOCOL

We train 50 independent models with seeds 42–91. For each seed s : train model M_s , split validation 50/50 into calibration $\mathcal{D}_{\text{cal}}^s$ and evaluation $\mathcal{D}_{\text{eval}}^s$, calibrate CP_s on $\mathcal{D}_{\text{cal}}^s$, evaluate on both $\mathcal{D}_{\text{eval}}^s$ and test. We report mean, 95% confidence intervals (t -distribution), and paired Wilcoxon tests (seed-level). This is *not* ensemble prediction—each seed is an independent experimental trial providing paired comparisons.

3.4 FEATURE OVERLAP METRIC

Feature temporal stability uses Jaccard similarity:

$$J(x_j) = \frac{|A_{\text{train}} \cap A_{\text{test}}|}{|A_{\text{train}} \cup A_{\text{test}}|} \quad (1)$$

where A_{train} and A_{test} are the sets of unique values for feature x_j in train and test data. In the pre-deployment framework (Section 7), we substitute validation data for test data as a proxy.

3.5 SHAP CONCENTRATION DIAGNOSTIC

We define the SHAP concentration metric as:

$$C = \frac{\phi_1}{\sum_{j=1}^p \phi_j} \quad (2)$$

where ϕ_j is the mean absolute SHAP value of the j -th feature computed on validation data using TreeExplainer [Lundberg et al., 2020], and ϕ_1 is the importance of the top-ranked feature. C is computed **before deployment** using only validation data.

3.6 WHY SHAP CONCENTRATION?

A natural question is whether simpler diagnostics suffice.

Feature overlap (Jaccard) is insufficient. Tasks relying on transaction IDs have Jaccard ≈ 0 while tasks with entity-based features have Jaccard > 0.5 (Table 2), yet Jaccard alone cannot explain the full range of coverage drops.

Post-hoc metrics require test data. Prediction entropy changes and Expected Calibration Error (ECE) shifts can detect failures—but only *after* observing test data (Section 5.4). Among the diagnostics we evaluate, SHAP concentration is available before deployment and shows the strongest severity association.

Why not raw feature importance? Raw importance values are not comparable across tasks (different scales, feature counts). Concentration normalizes to a $[0, 1]$ scale, enabling cross-task comparison.

Why top-1? We compared top-1 concentration against alternatives: top-2, top-3, Herfindahl-Hirschman Index (HHI), and feature importance entropy. Top-1 concentration is the only variant among these five that reaches nominal significance with coverage drop ($\rho = 0.833$, $p = 0.010$); all alternatives are non-significant ($p > 0.10$). Under Holm–Bonferroni correction for five metrics, the adjusted p -value for top-1 is 0.050 (boundary; Appendix G); the $n = 16$ cross-domain result ($\rho = 0.853$, $p < 0.001$) was not subject to this selection correction and provides independent replication. The discriminating power of top-1 arises because sales-office has high top-2/3 concentration (77–85%) but near-zero coverage drop—the diagnostic signal is specifically whether a *single* dominant feature drives predictions, not overall feature spread.

Why gradient-boosted models? This is not a scope limitation but a structural finding. TreeSHAP [Lundberg et al., 2018, 2020] computes exact Shapley values for all tree ensembles, including random forests. The distinction is architectural, not algorithmic. Random forests use bagging: each tree is trained independently on a bootstrap sample, so different trees may assign different top features; averaging across trees dilutes global single-feature concentration even

when individual trees concentrate heavily on one feature—confirmed empirically by RF $\rho = 0.30$ versus LightGBM $\rho = 0.833$. Gradient boosting’s sequential additive structure reinforces rather than dilutes single-feature dependence across rounds, so SHAP concentration faithfully measures a genuine model-level vulnerability. MLP-SHAP (kernel or deep variants) approximates rather than exactly decomposes contributions, adding noise that obscures structural concentration. The model-sensitivity analysis ($\rho = 0.833$ LightGBM, 0.667 CatBoost, 0.548 XGBoost vs. 0.30 RF, 0.43 MLP) thus identifies a boundary: *SHAP concentration is diagnostic where gradient boosting’s sequential structure produces genuine single-feature dependence*—a condition met by GBT architectures and not by bagging ensembles or neural networks.

4 THEORY: WHY CONCENTRATION RELATES TO FAILURE

We establish a mechanistic account for why SHAP concentration is associated with conformal failure under feature shift. The following theorem shows that when a model concentrates dependence on a single feature that shifts out of distribution, APS score and coverage bounds worsen monotonically with concentration parameter C , under explicit assumptions.

Notation. For a predicted distribution $\hat{p}(y | x) \in \Delta^{K-1}$, the APS conformity score $s(x, y^*)$ sums predicted probabilities from the most likely class down to and including y^* . Equivalently:

$$s(x, y^*) = 1 - \sum_{\substack{y \neq y^* \\ \hat{p}(y|x) < \hat{p}(y^*|x)}} \hat{p}(y | x), \quad (3)$$

since s includes all mass at or above the rank of y^* , leaving out only classes ranked strictly below. (The bound holds regardless of randomized tie-breaking conventions in APS implementations, since $\mathcal{B}$ uses strict inequality.)

Theorem 1 (Score Inflation Under Concentrated Feature Shift (Idealized Additive Model)). *Consider a K -class classifier ($K \geq 3$) with APS conformal predictor calibrated at level α with quantile $\hat{q}_\alpha$.² Let $C \in [0, 1]$ denote SHAP concentration on feature x_1 . Assume:*

(A1) **Additive decomposition.** $\hat{p}(y | x) = C \cdot g(y | x_1) + (1 - C) \cdot h(y | x_{\setminus 1})$, where $g(\cdot | x_1), h(\cdot | x_{\setminus 1}) \in \Delta^{K-1}$.³

²The theorem requires $K \geq 3$; the empirical endpoint additionally requires $K \geq 4$ to avoid the near-uniform score distribution that makes APS non-informative at $K = 3$ [Ding et al., 2023].

³Assumption (A1) posits additivity in probability space; TreeExplainer computes SHAP in *log-odds* space, so (A1) is an idealization. The theorem establishes directional intuition—monotone score inflation under concentrated dependence—rather than a for-

(A2) **Concentrated misclassification.** Under shift, $g(y^* | x_1^{\text{test}}) \leq \varepsilon$ for some $\varepsilon \in [0, 1/K)$.

(A3) **Residual exchangeability.** $(h(\cdot | x_{\setminus 1}^{\text{test}}), y_{\text{test}}^*) \stackrel{d}{=} (h(\cdot | x_{\setminus 1}^{\text{cal}}), y_{\text{cal}}^*)$.

Then:

(i) **Pointwise score bound.** For each test instance (x^{test}, y^*) :

$$s(x^{\text{test}}, y^*) \geq 1 - (K-1)[C\varepsilon + (1-C)h(y^* | x_{\setminus 1}^{\text{test}})]. \quad (4)$$

(ii) **Expected score inflation.** Let $\bar{h} := \mathbb{E}[h(y_{\text{cal}}^* | x_{\setminus 1}^{\text{cal}})]$ denote the expected true-class probability under the residual model. Then:

$$\mathbb{E}[s_{\text{test}}] \geq C(1 - (K-1)\varepsilon) + (1-C)(1 - (K-1)\bar{h}). \quad (5)$$

(iii) **Monotone vulnerability.** The bound in (5) is strictly increasing in C whenever $\varepsilon < \bar{h}$. A sufficient condition is $\bar{h} \geq 1/K$ (the residual model is at least uniform on the true class); together with (A2) this implies $\varepsilon < \bar{h}$.

(iv) **Coverage degradation bound.** Coverage satisfies:

$$\mathbb{P}(y^* \in \mathcal{C}(x^{\text{test}})) \leq \mathbb{P}(h(y_{\text{cal}}^*) \geq T(C)), \quad (6)$$

where $T(C) := [(1 - \hat{q}_\alpha)/(K-1) - C\varepsilon]/(1-C)$ is strictly increasing in C when $\varepsilon < (1 - \hat{q}_\alpha)/(K-1)$ (satisfied at $\varepsilon = 0$). Hence the coverage upper bound is non-increasing in C .

Proof sketch. Part (i) upper-bounds omitted APS mass by $(K-1)\hat{p}(y^*)$ and substitutes assumptions (A1)–(A2). Part (ii) takes expectations and uses residual exchangeability (A3). Part (iii) differentiates the lower bound in (5) and gives monotonicity when $\varepsilon < \bar{h}$. Part (iv) rearranges (4) to obtain (6) via threshold $T(C)$ and shows $T'(C) > 0$ under the stated condition. $\square$

The full derivation and conservative bound verification are in Appendix H. Empirically, catastrophic tasks show strong rightward conformity-score shifts (KS = 0.68–0.96), while concentration and shift magnitude are weakly related across SALT ($\rho = 0.29$, $p = 0.49$). This supports interpreting concentration as a model-vulnerability proxy (dependence structure) rather than a shift-size metric.

5 RESULTS

5.1 COVERAGE DEGRADATION ACROSS TASKS

Despite experiencing identical COVID-19 temporal shift, the 8 supply chain tasks exhibit dramatically different conformal guarantee for the measured SHAP-derived C . The empirical proxy is validated by $\rho = 0.853$ and conservative-bound verification (Appendix H).

erage degradation (Table 1). All coverage drops are statistically significant (paired Wilcoxon $p \leq 0.005$, 95% CIs from the t -distribution), yet range from 0.1% to 77.1%.

5.2 DIAGNOSTIC ANALYSIS

Two factors account for much of the variance:

Factor 1: Feature temporal stability. Tasks using transaction IDs (Jaccard ≈ 0.02) fail catastrophically; tasks using stable entities (products, organizations; Jaccard > 0.5) maintain coverage (Table 2).

Factor 2: Feature importance concentration. Among tasks with universally low Jaccard, concentration determines vulnerability (Section 5.3).

5.3 SHAP CONCENTRATION CORRELATES WITH FAILURE

Across all 8 SALT multiclass tasks, SHAP concentration correlates with coverage degradation: Spearman $\rho = 0.833$, $p = 0.010$ (Table 3, Figure 2). Bootstrap 95% CI: $[0.30, 1.00]$ (10,000 resamples, percentile method; BCa would yield a modestly tighter lower bound at $n = 8$ but does not change the qualitative conclusion). The correlation magnitude is stable under leave-one-out analysis ($\rho \in [0.75, 0.96]$), though statistical significance is not maintained for 2 of 8 jackknife samples ($p = 0.052$) due to reduced power at $n = 7$. SHAP concentration values are highly stable: bootstrap resampling (1,000 resamples of 10K validation samples) yields coefficient of variation below 1% for all tasks, with 95% CIs within ± 1 percentage point.

Mechanism. Figure 1 reveals the distinction. Catastrophic tasks: the dominant feature experiences a concentrated importance explosion ($4.5\times$ increase). Robust tasks: importance redistributes across features with ranking reshuffle ($10\text{--}20\times$ increases distributed). This matches the prediction of Theorem 1.

False positives. Two tasks with concentration $> 40\%$ remain robust: sales-office (secondary feature SALESORG has Jaccard = 0.61) and item-shippoint (high model variance). These motivate the protective-factor check in our framework (Section 7). The Jaccard > 0.5 and importance $> 15\%$ protective-factor rule is derived from $n = 1$ observation (sales-office) and should be treated as provisional until validated on additional false-positive cases.

Task independence. Intraclass correlation analysis confirms task identity explains the majority of seed-level variance in coverage drops (ICC = 0.675, 95% CI $[0.47, 0.90]$; effective $n_{\text{eff}} = 11.7$). Per-task ICC(val,test) is negative for all 8 tasks, indicating paired tests are conservative. Details in Appendix G.

Statistical power. At $n = 8$, power to detect $\rho = 0.833$ is

≈ 0.76 (below the 0.80 convention); the within-SALT result is therefore *exploratory*. The $n = 16$ cross-domain result ($\rho = 0.853$, $p < 0.001$; power > 0.99) is the confirmatory endpoint.

Cross-domain extension. Extending to 16 multiclass tasks across 9 domains—including 8 external datasets spanning documented geographic, temporal, and benchmark-partition shifts (Section 6.5; shift mechanisms for 5 UCI pre-defined-split datasets are not separately documented)—yields $\rho = 0.853$, $p < 0.001$ (Kendall $\tau = 0.667$; bootstrap 95% CI $[0.50, 0.96]$). We note that the 40% threshold was derived from the SALT subset ($n = 8$); the $n = 16$ correlation therefore includes the in-sample SALT tasks ($\rho = 0.833$ within SALT alone) alongside the held-out external 8 datasets on which the threshold was not tuned. The external datasets provide the cleanest out-of-sample validation: Covertypes is correctly flagged as catastrophic (81.8 pp drop, 10/10 seeds), 6 deterministic low-concentration datasets remain robust (10/10 seeds each), and KDDCup99 is the principal intermediate-regime false negative.

5.4 DIAGNOSTIC COMPARISON

We compare SHAP concentration against two alternative pre-deployment diagnostics that require no test data (Table 4).

Native feature importance concentration: top-1 LightGBM gain divided by total gain (zero SHAP cost, single seed; seed=42). Achieves $\rho = 0.667$ ($p = 0.071$)—directionally correct but not significant at $\alpha = 0.05$. The weaker discrimination arises because native gain conflates feature cardinality with predictive importance: sales-office shows 66.1% native concentration (vs. 42.6% SHAP) despite near-zero coverage drop.

Ensemble disagreement: standard deviation of validation coverage across 50 seeds. Achieves $\rho = 0.452$ ($p = 0.260$). Disagreement captures model instability but not the feature-level mechanism driving conformal failure.

Class cardinality as confound. Within SALT, $\log(\text{num_classes})$ alone correlates with coverage drop ($\rho = 0.743$, $p = 0.035$), and both partial correlations are non-significant at $n = 8$ ($\rho_{\text{partial}} = 0.629$, $p = 0.131$ for concentration; 0.334 , $p = 0.464$ for cardinality; Table 4). This non-significance is expected at $n = 8$ —within-SALT the two predictors are correlated and power is insufficient for causal separation. Cross-domain evidence provides complementary support: Covertypes (7 classes, $C = 49.8\%$) fails catastrophically, while higher-cardinality low-concentration datasets such as Avila (12 classes, $C = 20.5\%$), PAMAP2 (12, 19.2%), and Pendigits (10, 14.5%) remain robust—showing that high class count alone does not predict failure. KDDCup99 (11, 21.1%) is a seed-dependent intermediate case rather than a clean

Table 1: Coverage Degradation Under COVID-19 Distribution Shift (50 Seeds). 95% CIs from t -distribution. p -values from paired Wilcoxon signed-rank test (val $\rightarrow$ test, seed-level). All drops significant at $p \leq 0.005$.

Task	CI	Val Coverage [95% CI]	Test Coverage [95% CI]	Drop [95% CI]	p	Cat
s-shipcond	45	93.5 [93.2, 93.7]	21.8 [14.1, 29.6]	71.6 [63.9, 79.4]	<0.001	SEV
s-group	459	83.6 [81.7, 85.5]	12.4 [3.1, 21.7]	71.2 [61.3, 81.0]	<0.001	SEV*
s-payterms	137	90.8 [90.7, 91.0]	13.7 [6.0, 21.5]	77.1 [69.4, 84.9]	<0.001	SEV
i-plant	35	92.0 [91.7, 92.3]	81.4 [79.0, 83.8]	10.6 [8.2, 13.0]	<0.001	ROB
i-shippoint	69	91.2 [90.9, 91.4]	72.7 [62.3, 83.0]	18.5 [8.2, 28.8]	0.005	ROB* [†]
s-incoterms	13	95.5 [95.3, 95.7]	87.0 [84.7, 89.3]	8.5 [6.2, 10.8]	<0.001	ROB
i-incoterms	13	95.0 [94.9, 95.2]	83.7 [80.9, 86.6]	11.3 [8.5, 14.1]	<0.001	ROB
s-office	25	99.9 [99.9, 100.0]	99.9 [99.9, 99.9]	0.1 [0.0, 0.1]	<0.001	ROB

SEV = Severe (>50% drop), ROB = Robust (<20% drop); *high model variance (CV > 50%), classified by median; s-group’s 83.6% validation coverage is below the 90% target—sub-nominal non-randomized APS coverage at $K = 459$ is expected [Ding et al., 2023]; [†]meets At-risk criterion (drop > 15 pp) under the decision framework (Table 8).

Table 2: Feature Overlap for Primary Features.

Task	Primary Feature	Jaccard	Drop
s-shipcond	SALESDOC (ID)	0.02	71.6%
s-group	SALESDOC (ID)	0.02	71.2%
i-incoterms	PRODUCT, PARTY	0.58	11.3%
s-office	SALESORG	0.61	0.1%

Drop = mean val – mean test coverage (50 seeds).

robust counterexample. At $n = 16$ (8 SALT + 8 external multiclass tasks), the partial Spearman controlling for $\log K$ yields $\rho_{\text{partial}}(\text{conc.}) = 0.771$ ($p = 0.0008$) while $\rho_{\text{partial}}(\log K) = -0.010$ ($p = 0.97$), providing strong cross-domain evidence that concentration drives the severity association independently of class count.

We additionally compare against two *post-hoc* baselines computed from 10-seed ACI experiments:

Prediction entropy change: $\Delta H = H_{\text{test}} - H_{\text{val}}$, where $H = -\sum_c p_c \log p_c$ averaged over instances.

Expected Calibration Error change: $\Delta \text{ECE} = \text{ECE}_{\text{test}} - \text{ECE}_{\text{val}}$.

Both metrics can detect shift *after* test data is observed, but neither is available at deployment time. SHAP concentration is computed solely on validation data, making it actionable for pre-deployment risk assessment. Comparison across 7 tasks in Appendix F.

6 EXTENDED EXPERIMENTS

6.1 ADAPTIVE CONFORMAL INFERENCE ACROSS ALL TASKS

ACI [Gibbs and Candès, 2021] recovers marginal coverage on severe tasks (+47.5 pp on s-shipcond, +71.8 pp on s-group, +60.0 pp on s-payterms) but inflates prediction sets; only s-shipcond remains deployable post-adaptation (usable-

Table 3: Stratified Correlation Analysis.

Group	n	ρ	Kendall τ	$p(\rho)$	Boot. 95% CI
Multiclass (SALT)	8	0.833	0.714	0.010	[0.30, 1.00]
Multiclass (4 dom.)	11	0.909 [†]	0.782	<0.001	[0.61, 1.00]
Multiclass (8 dom.)	15	0.882 [‡]	0.714	<0.001	[0.60, 0.97]
Multiclass (9 dom.)	16	0.853	0.667	<0.001	[0.50, 0.96]
Combined (10 dom.) [§]	17	0.654	—	0.004	—

LOO stability (SALT): $\rho \in [0.75, 0.96]$; 6/8 jackknife significant.

Multiclass (9 domains) is the primary result, excluding binary ceiling-effect tasks. Under the at-risk criterion (drop > 15 pp), threshold performance at 40% is Precision = 0.83, Recall = 0.83 (Appendix D). [†]Single-seed external values; multi-seed-consistent value at $n = 11$ is $\rho = 0.818$, $p = 0.002$ (see Appendix G).

[‡] $n = 15$ is an intermediate analysis (7 external multiclass domains); $n = 16$ adds KDDCup99 as the 8th external domain to complete the 9-domain primary endpoint. Stack Overflow (3 classes, near-binary ceiling effect) is excluded from all multiclass endpoints. [§]Combined row includes all 17 tasks (8 SALT + 9 external, 10 domains: SALT + 9 external), adding Stack Overflow back to the $n = 16$ primary endpoint. Spearman $\rho = 0.654$, $p = 0.004$ (from Appendix G); Kendall τ not reported for this auxiliary endpoint.

set rate 30%: sets $\approx 30\%$ of $K=45$ classes; s-group inflates to 74–89% of 459). Robust tasks show minimal benefit. ACI can recover coverage but often sacrifices informativeness in the highest-risk regimes.

6.2 ALTERNATIVE CONFORMAL SCORING

Regularized Adaptive Prediction Sets (RAPS) [Angelopoulos et al., 2021] shows mechanism-dependent benefit: substantially reduces drops for high-cardinality tasks (s-group: 73.5% $\rightarrow$ 10.4%; s-payterms: 79.9% $\rightarrow$ 35.1%) but not the most concentrated task (s-shipcond: 60.4% $\rightarrow$ 67.8%). This matches Section 4: regularization controls class-accumulation effects, not concentrated single-feature dependence (Appendix J; RAPS used random 50/50 calibration splits vs. deterministic main-experiment splits—within-

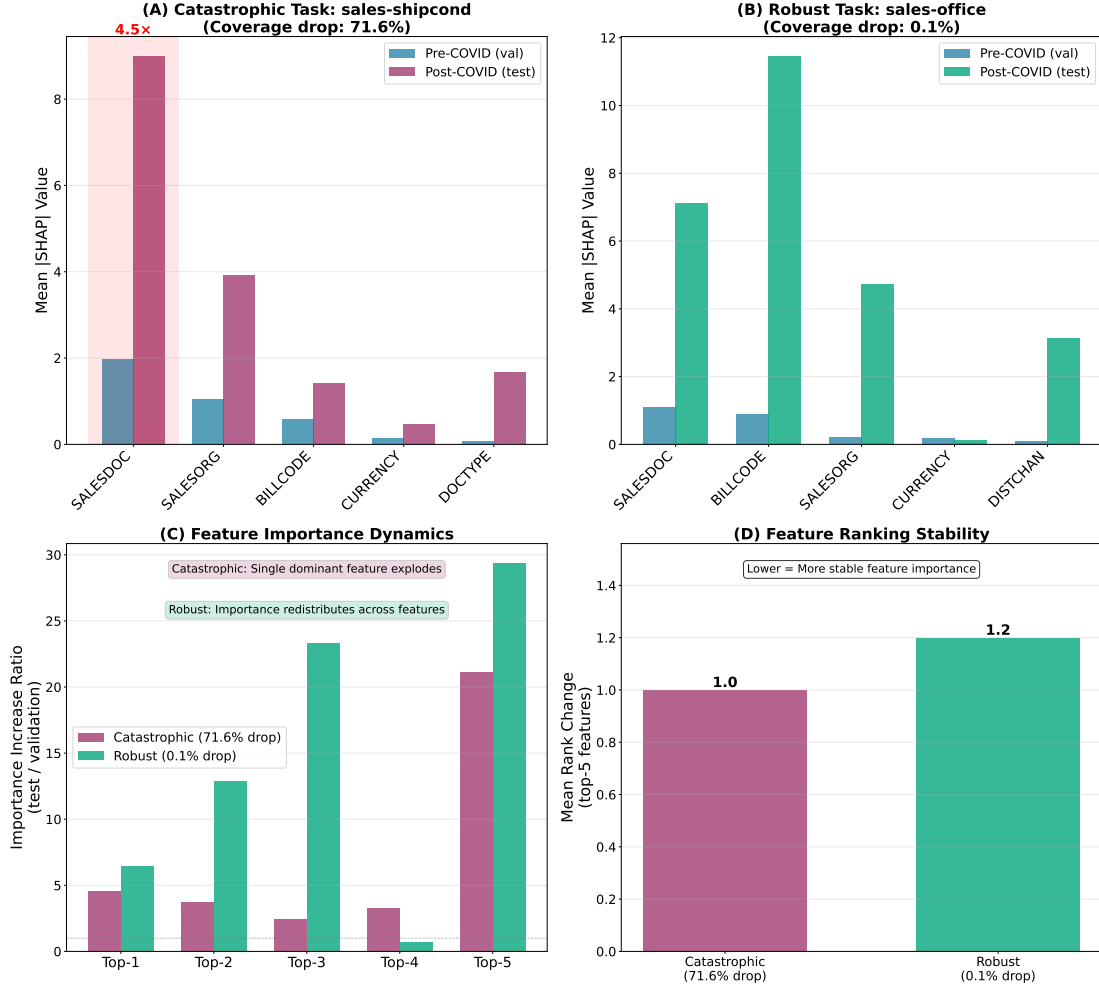


Figure 1: **Feature Importance Dynamics Under Shift.** (A) Catastrophic task: dominant feature explodes $4.5\times$ while maintaining top rank (concentrated dependence). (B) Robust task: importance redistributes across features with ranking reshuffle (diversified dependence). (C–D) Importance dynamics and ranking stability. Despite $\sim 0\%$ Jaccard similarity, coverage drops range from negligible to catastrophic.

Table 4: Pre-Deployment Diagnostic Comparison ($n = 8$). Spearman ρ with 50-seed mean coverage drop. ρ_{partial} : partial Spearman controlling for the other variable in the SHAP–class-count pair.

Diagnostic	ρ	p	ρ_{partial}	p_{partial}
SHAP concentration	0.833	0.010	0.629	0.131
$\log(\text{num_classes})$	0.743	0.035	0.334	0.464
Native FI concentration	0.667	0.071	—	—
Ensemble disagreement	0.452	0.260	—	—

6.3 COMPARISON WITH STANDARD SHIFT DETECTION

MMD, C2ST, and PSI detect shift on all 8 SALT tasks (MMD $p < 0.002$, C2ST $> 99.9\%$, PSI > 0.002) but show $\rho \leq 0.19$ with coverage drop—confirming shift detection does not predict failure severity.

6.4 RETRAINING ANALYSIS

Quarterly retraining improves s-shipcond by +18.9 pp ($p = 0.036$, unadjusted, single-seed; Holm-corrected $p = 0.11$ over 3 severe tasks), partially improves s-payterms (+15.5 pp, non-significant), and is ineffective for s-group ($0.0\% \rightarrow 2.2\%$). Improvement concentrates in the final two months of the window; robust tasks remain near-ceiling without intervention.

experiment comparisons are unaffected).

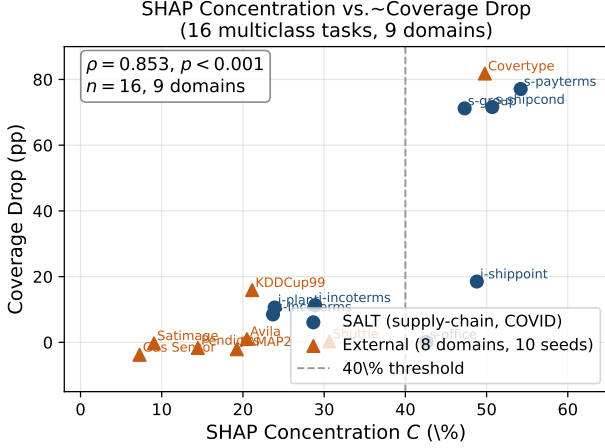


Figure 2: **SHAP Concentration Correlates with Coverage Degradation (Primary Result).** Spearman $\rho = 0.853$, $p < 0.001$ across all 16 multiclass tasks in 9 domains (dark circles: 8 SALT supply-chain tasks; orange triangles: 8 external domains, 10-seed means). The 40% threshold (dashed) is derived from validation data only. Within SALT alone: $\rho = 0.833$, $p = 0.010$ ($n = 8$).

6.5 CROSS-DOMAIN AND EXTERNAL VALIDATION

External validation on 9 non-supply-chain datasets [Dua and Graff, 2017] with the same pipeline (Appendix A.6) replicates the pattern: $\rho = 0.853$, $p < 0.001$ (Kendall $\tau = 0.667$) across $n = 16$ multiclass tasks. The 8 external multiclass datasets comprise 4 with documented real-world shifts (Covertypes geographic, Gas Sensor temporal, KDDCup99 attack-distribution, PAMAP2 subject-based) and 4 null-shift controls (Shuttle, Avila, Pendigits, Satimage: random/pre-defined splits, no genuine shift). The controls are informative: the framework predicts robust coverage for low-concentration models absent genuine shift, confirmed in all 4 cases (10/10 seeds each). Among documented-shift datasets, Covertypes ($C = 49.8\%$, 81.8 pp drop, 10/10 seeds) is the key catastrophic external case; KDDCup99 ($C = 21.1\%$, drop = 15.9 ± 21.4 pp) is the principal intermediate-regime false negative.

6.6 PLACEBO TEST

A pre-COVID temporal placebo (train 2018, validate 2019-H1, test 2019-H2) shows $6\text{--}143\times$ lower degradation than the COVID split (Appendix B), confirming the diagnostic is most informative under severe event-driven shift; the 1–2 order-of-magnitude contrast holds despite the asymmetric training window.

7 DECISION FRAMEWORK

This exploratory protocol guides practitioners deploying conformal prediction under potential shift:

1. **Compute** $C = \phi_1 / \sum_j \phi_j$ (Eq. 2). If $C \leq 40\%$: lower observed risk. If $C > 40\%$: check protective factors—secondary features with train-validation Jaccard > 0.5 and importance $> 15\%$ (thresholds empirically derived from a single observed case; treat as provisional); absent these, flag as **vulnerable**. If $C \in [35\%, 45\%]$: treat as **uncertain risk** with monitoring regardless of other factors.
2. **If vulnerable**: consider quarterly retraining (suggestive effect; Holm-corrected $p = 0.11$; single-seed; Appendix E).
3. **Monitor** empirical coverage continuously; defer retraining for lower-risk tasks.

8 CONCLUSION

Catastrophic tasks become *confidently wrong*—prediction sets shrink (rightward score shift, Appendix I) while coverage collapses, driven by concentrated OOD-feature dependence that generic shift detectors cannot distinguish from robust outcomes. SHAP concentration provides a pre-deployment signal: $\rho = 0.853$ ($p < 0.001$) across 16 multiclass tasks vs. $\rho \leq 0.19$ for MMD/C2ST. Mitigation is mechanism-dependent (RAPS helps class-accumulation failures; retraining/ACI recover coverage). The 40% threshold is exploratory ($n = 8$ SALT tasks) and requires prospective validation beyond boosting models.

Author Contributions

C.L. conceived the research question, designed and executed all experiments, performed the statistical analysis, and wrote the paper.

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Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study (Supplementary Material)

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A REPRODUCIBILITY DETAILS

All experimental code—including SHAP concentration computation, conformal prediction evaluation, statistical tests, and figure generation—is submitted as supplementary material.

A.1 LIGHTGBM HYPERPARAMETERS

All models trained with fixed LightGBM [Ke et al., 2017] settings: objective=multiclass, boosting=gbd, num_leaves=31, learning_rate=0.05, feature_fraction=0.8, bagging_fraction=0.8, bagging_freq=5, num_boost_round=500, early_stopping_rounds=50, seeds 42–91 (50 seeds). No task-specific tuning was performed.

A.2 CONFORMAL PREDICTION

Method: APS [Romano et al., 2020]. Target coverage: $(1 - \alpha) = 0.9$ (90%). Calibration: deterministic first-half/second-half split of the validation set (i.e., the first $\lfloor n_{\text{val}}/2 \rfloor$ records form the calibration set). This preserves temporal order within the validation period but is not expected to affect the rank-order concentration–drop correlation, as the same split is applied uniformly across all tasks. Quantile: $q = \min(\lceil (n + 1)(1 - \alpha) \rceil / n, 1)$.

Calibration and test set sizes. Sales-level tasks (s-shipcond, s-payterms, s-group, s-incoterms, s-office): $n_{\text{val}} = 71,474$, $n_{\text{cal}} = 35,737$, $n_{\text{test}} = 88,942$. Item-level tasks (i-plant, i-shippoint, i-incoterms): $n_{\text{val}} = 293,896$, $n_{\text{cal}} = 146,948$, $n_{\text{test}} = 402,855$. At these calibration set sizes, the finite-sample correction is negligible: $q - (1 - \alpha) < 3 \times 10^{-5}$ for all tasks, so finite-sample effects do not contribute to the observed coverage gaps.

A.3 SHAP COMPUTATION

Method: TreeExplainer [Lundberg et al., 2020]. Subsample: 10,000 per split. Aggregation: mean absolute SHAP across classes. Concentration metric (Eq. 2): computed on validation data only.

A.4 BASELINE DIAGNOSTICS

Native feature importance concentration uses a single model (seed=42) with LightGBM’s gain-based importance; concentration is top-1 gain divided by total gain. Ensemble disagreement is the standard deviation of validation coverage across 50 seeds. Both are computed without SHAP.

A.5 COMPUTATIONAL RESOURCES

Standard CPU (8 cores, 8GB RAM). Wall-clock: ~ 3 –4 hours for full 8-task, 50-seed suite. Software: Python 3.9, LightGBM 3.3, SHAP 0.41.

A.6 EXTERNAL DATASET PROTOCOLS

Table 5 documents the train/test split, shift mechanism, and dataset characteristics for all 9 external validation datasets. All use the same LightGBM+APS pipeline as SALT (Appendix A), with 10 seeds and no hyperparameter tuning.

Table 5: External Dataset Protocols. Cal = calibration set (50% of train); n_{tr} , n_{te} = approximate train/test sizes used; Seeds = 10 for all external datasets. Type: **DS** = documented real-world shift; **NC** = null-shift control (random/pre-defined split, no genuine shift; low concentration and robust coverage are the expected and observed outcome).

Dataset	Shift mechanism	Type	K	n_{tr}	n_{te}	Source
Covertypes	Geographic: areas 1–2 (train), area 3 (val/cal), area 4 (test)	DS	7	290,680	36,968	[Dua and Graff, 2017]
Gas Sensor	Temporal batch drift: batches 1–6 (train), 7–8 (cal), 9–10 (test)	DS	6	9,840	4,070	[Dua and Graff, 2017]
KDDCup99	Attack-type distribution: official KDD Cup 1999 train/test files	DS	11	395,073	98,769	[Dua and Graff, 2017]
PAMAP2	Population shift: subject-based split (unseen subjects at test)	DS	12	1,674,379	268,493	[Dua and Graff, 2017]
Shuttle	Null-shift control: UCI Statlog; 60/20/20 random split	NC	7	46,400	11,600	[Dua and Graff, 2017]
Avila Bible	Null-shift control: copyist identification; 50/50 random split	NC	12	10,430	10,437	[Dua and Graff, 2017]
Pendigits	Null-shift control: UCI pre-defined split (digit recognition)	NC	10	7,494	3,498	[Dua and Graff, 2017]
Satimage	Null-shift control: satellite image; 60/20/20 random split	NC	6	5,144	1,286	[Dua and Graff, 2017]
Stack Overflow	Category shift (3-class subset); excluded from $n = 16$ primary endpoint (near-binary ceiling effect)	—	3	65,105	127,370	[Dua and Graff, 2017]

For Covertypes, train uses wilderness areas 1–2 (Spruce-Fir and Lodgepole Pine); conformal calibration uses area 3 (Ponderosa Pine); test uses area 4 (Cottonwood/Willow). For Gas Sensor, each batch contains $\sim 1,000$ samples across 6 gas types; temporal sensor drift is the documented shift mechanism; n_{tr} is batches 1–8 (train + conformal cal); n_{te} is batches 9–10. For KDDCup99, classes are the 11 attack categories present in the 10% subset; the official train file ($\sim 395K$) is split 75/25 into LightGBM training and conformal calibration; n_{tr} reports total official training file; n_{te} is the official test file. For PAMAP2, a subject-based protocol yields a large training pool ($\sim 1.67M$ samples); n_{tr} = LightGBM train + conformal cal combined. For Pendigits, the official UCI train partition (7,494) is split 80/20 into LightGBM training and conformal calibration. For Shuttle, Avila, Satimage, and Stack Overflow, the full dataset is split 60/20/20 into LightGBM train/conformal cal/test; n_{tr} = train+cal pool. SHAP concentration is computed on the conformal calibration partition (validation data) only. Null-shift controls (NC) serve as low-concentration baselines: the framework predicts robust coverage absent genuine shift, which is confirmed in all 4 cases (10/10 seeds each).

B PLACEBO TEST DETAILS

C HIGH MODEL VARIANCE ANALYSIS

Tasks marked with * in Table 1 exhibit extreme model variance ($CV > 50\%$):

- **s-group**: Mean 12.4%, median **0.5%**. Most models fail; outliers inflate mean.
- **i-shipoint**: Mean 72.7%, median **89.9%**. Most models succeed; failures deflate mean.

Table 6: Placebo Test: Pre-COVID vs COVID Degradation (50-Seed Mean Drops).

Task	Placebo	COVID	Ratio
s-shipcond	0.5%	71.6%	143×
s-group	2.0%	71.2%	36×
s-payterms	0.0%	77.1%	>100×
i-plant	1.8%	10.6%	6×
i-shippoint	1.5%	18.5%	12×
s-incoterms	1.2%	8.5%	7×
i-incoterms	0.6%	11.3%	19×
s-office	0.1%	0.1%	1×

D THRESHOLD SENSITIVITY ANALYSIS

Table 7 shows threshold performance across $n = 16$ multiclass tasks. Note that the 40% threshold is derived from the SALT concentration gap (in-sample); the 8 external datasets are fully out-of-sample. On the external subset alone at 40%: Precision = 1.00, Recall = 0.50 (1 TP: Covertypes; 1 FN: KDDCup99; 6 TN). Low external recall reflects sparse catastrophic cases; the primary claim rests on the $n = 16$ correlation ($\rho = 0.853$) rather than threshold classification. Ground-truth labels use the at-risk rule (mean coverage drop >15 pp, consistent with 50-seed means in Table 1). For high-variance tasks, the mean and median labels can differ: i-shippoint has mean drop 18.5 pp (At-risk) but median drop 1.2 pp (robust); Table 1 labels it ROB by median for descriptive purposes, while the framework uses the mean-based threshold for threshold evaluation.

Table 7: Threshold Sensitivity for SHAP Concentration Across 16 Multiclass Tasks (8 SALT + 8 external; Step 2 only, no protective-factor check). At-risk defined as coverage drop >15%.

Threshold	Prec	Rec	F1	TP	FP	FN
25%	0.63	0.83	0.71	5	3	1
30%	0.71	0.83	0.77	5	2	1
35%	0.83	0.83	0.83	5	1	1
40%	0.83	0.83	0.83	5	1	1
45%	1.00	0.83	0.91	5	0	1
50%	1.00	0.33	0.50	2	0	4

At 40%, the single FP is s-office (concentration 42.6%) which has a protective secondary feature (SALESORG Jaccard = 0.61), and the single FN is KDDCup99 (intermediate regime; $C = 21.1\%$, drop = 15.9 pp). At 30%, the two FPs are s-office and Shuttle ($C = 30.7\%$); at 25%, i-incoterms ($C = 28.9\%$) is additionally flagged. Applying the protective-factor check at 40% removes s-office (precision = 1.00) but leaves the KDDCup99 FN (recall = 0.83).

E DECISION FRAMEWORK VALIDATION

Seed stability protocol. KDDCup99’s multi-seed analysis ($C = 21.1 \pm 7.5\%$) reveals that single-seed SHAP concentration can be unstable. We recommend computing C over $k \geq 5$ seeds and using the mean. Tasks in the ambiguous zone (35–45%) should receive additional scrutiny: if the upper 95% CI exceeds 40%, treat as potentially vulnerable. This safety override is designed to reduce KDDCup99-like false negatives in intermediate regimes. For SALT tasks, concentration is stable across 50 seeds (within-task CV < 5%); instability arises primarily when the top-feature identity varies across seeds.

Threshold selection. The 40% threshold is proposed based on the empirical distribution of SHAP concentration values on validation data: a natural gap separates low-concentration tasks (24–29%) from high-concentration tasks (43–54%), visible in Table 8. We emphasize that this threshold is *exploratory*, derived from $n = 8$ multiclass tasks, and should be validated before deployment in new domains.

Cross-domain transfer test. The 40% threshold applied without tuning to 9 external domains yields 7/9 deterministic or near-deterministic outcomes (6 at 10/10, Shuttle at 9/10), with Covertypes correctly flagged as catastrophic and Gas Sensor correctly unflagged as robust. KDDCup99 is the principal intermediate-regime false negative, while Stack Overflow (3

Table 8: Concentration Check Applied to All Multiclass Tasks (Actual = At-risk if drop >15 pp).

Task	Conc. (%)	Step 2	Actual (>15 pp)	Note
<i>SALT (in-sample)</i>				
s-payterms	54.2	VULN	At-risk	—
s-shipcond	50.7	VULN	At-risk	—
i-shippoint	48.8	VULN	At-risk*	High variance [†]
s-group	47.3	VULN	At-risk*	—
s-office	42.6	VULN	ROB	Protective [‡]
i-incoterms	28.9	ROB	ROB	—
i-plant	23.9	ROB	ROB	—
s-incoterms	23.7	ROB	ROB	—
<i>External (held-out)</i>				
Covertime	49.8	VULN	At-risk	10/10 seeds
Shuttle	30.7	ROB	ROB	9/10 seeds
Avila Bible	20.5	ROB	ROB	10/10 seeds
PAMAP2	19.2	ROB	ROB	10/10 seeds
KDDCup99	21.1	ROB	At-risk [§]	Seed-dependent (FN)
Pendigits	14.5	ROB	ROB	10/10 seeds
Satimage	9.0	ROB	ROB	10/10 seeds
Gas Sensor	7.3	ROB	ROB	10/10 seeds
Stack Overflow	48.9	VULN	ROB	Near-binary ceiling [¶]

[†]Median drop 1.2%; most seeds maintain coverage. [‡]SALESORGANIZATION: train-validation Jaccard=0.61, importance>15%.

[§]Concentration below threshold ($C = 21.1 \pm 7.5\%$), but mean drop exceeds the at-risk cutoff (15.9 pp); outcome seed-dependent (range −0.8 to 73.5 pp). In-sample F1 = 0.89 (1 FP: s-office). External: Covertime deterministic catastrophic (10/10); five low-concentration datasets deterministic robust (10/10); Shuttle near-deterministic (9/10); KDDCup99 is the intermediate-regime false negative. [¶]Stack Overflow ($K = 3$, $C = 48.9\%$): high concentration predicts VULN, but near-binary class structure creates a ceiling effect (only 3 classes; APS prediction sets trivially remain small). Actual coverage is robust (−7.0 pp). Excluded from $n = 16$ multiclass primary endpoint; confirms diagnostic applies to multiclass ($K > 3$) settings only.

classes) exhibits the near-binary ceiling effect. Across the $n = 16$ multiclass set: $\rho = 0.853$, $p < 0.001$ (Section 6.5). The sensitivity analysis in Appendix D reports threshold performance across 30–50%; given sparse catastrophic external cases, this threshold should be treated as provisional.

F PRE-DEPLOYMENT VS POST-HOC BASELINES

SHAP concentration is computed on validation data *before* deployment. Prediction entropy and ECE require observing test data. This section reports entropy and ECE changes for completeness, noting that they are not available as pre-deployment diagnostics.

Table 9: Pre-Deployment (SHAP) vs Post-Hoc (Entropy, ECE) Shift Diagnostics (10-Seed ACI Experiments). SHAP concentration is computed on validation data only. Δ Entropy and Δ ECE require test-time observations.

Task	Cat	SHAP Conc. [†]	Δ Entropy	Δ ECE
<i>Severe tasks</i>				
s-shipcond	SEV	50.7%	−0.746	+0.408
s-group	SEV	47.3%	−1.274	+0.176
s-payterms	SEV	54.2%	−1.979	+0.629
<i>Robust tasks</i>				
i-plant	ROB	23.9%	+0.181	+0.131
i-shippoint	ROB	48.8%	+0.469	+0.073
s-incoterms	ROB	23.7%	+0.364	+0.350
s-office	ROB	42.6%	+0.032	+0.104

[†]Pre-deployment diagnostic (computed on validation data only). item-incoterms omitted (3-seed pilot only).

Counter-intuitively, all three catastrophic tasks (coverage drop $> 70\%$) show *decreasing* prediction entropy ($\Delta H < 0$): the model becomes *more* confident at test time, not less. This occurs because concentrated dependence on an OOD feature produces arbitrary but confident predictions. In contrast, the two robust tasks with non-trivial drops (i-plant, i-shippoint; coverage drops of 10–19%) show *increasing* entropy ($\Delta H > 0$), consistent with standard distributional broadening. A practitioner monitoring only entropy would correctly detect shift in moderate cases but would be *misled* in the most damaging catastrophic cases—observing decreasing uncertainty while coverage collapses. ECE increases across all tasks but does not discriminate between severity levels (both catastrophic and robust tasks show increases of +0.07 to +0.63). In this SALT comparison, SHAP concentration is the only evaluated pre-deployment metric that flags all three catastrophic tasks.

G TASK INDEPENDENCE ANALYSIS

We analyze pseudo-replication in the 50-seed design using intraclass correlation (ICC).

Between-task ICC. For coverage drops, $\text{ICC} = 0.675$ (95% CI: [0.47, 0.90]; $F(7, 392) = 104.6$), confirming that task identity explains the majority of seed-level variance. The design effect is 34.1, yielding effective $n_{\text{eff}} = 400/34.1 = 11.7$ —exceeding the 8 task-level observations. For test coverages, $\text{ICC} = 0.720$ (95% CI: [0.52, 0.92]). These values confirm that the $n = 8$ tasks constitute meaningfully independent observations for the Spearman correlation analysis.

Per-task ICC(val,test). For each task, we compute ICC between the 50 validation coverages and 50 test coverages (paired by seed). All 8 per-task ICCs are negative (−0.80 to −0.10), indicating that seeds producing above-average validation coverage do *not* produce above-average test coverage. This means paired Wilcoxon tests are conservative: the effective sample size for within-task paired tests *exceeds* 50, so the reported p -values are, if anything, too large.

Multiple comparisons. We tested 5 concentration metrics (top-1, top-2, top-3, HHI, entropy) against coverage drop. Under Holm–Bonferroni correction for this 5-test family, the adjusted p -value for top-1 concentration is $5 \times 0.010 = 0.050$, at the conventional significance boundary. For the primary correlation subsets ($n = 8$, $n = 11$, $n = 16$), all adjusted p -values remain significant (Holm-corrected $p \leq 0.021$). We note that top-1 was the a priori natural choice (Section 3.6), not one of 5 equally plausible alternatives; the correction is therefore conservative.

Multi-seed aggregation. The primary $n = 16$ correlation ($\rho = 0.853$) uses 10-seed means for all external datasets and 50-seed means for SALT tasks. The early $n = 11$ result ($\rho = 0.909$) used single-seed external values; switching to

Table 10: Per-Task ICC Between Validation and Test Coverages (50 Seeds).

Task	ICC(val,test)	Design Effect	Eff. n
s-payterms	−0.80	0.20	250
s-shipcond	−0.77	0.23	220
s-group	−0.72	0.28	177
i-plant	−0.42	0.58	86
i-incoterms	−0.37	0.63	80
s-incoterms	−0.35	0.65	77
s-office	−0.15	0.85	59
i-shippoint	−0.10	0.90	55

multi-seed means at $n = 11$ yields $\rho = 0.818$, $p = 0.002$, with the reduction driven by KDDCup99 whose multi-seed mean drop (15.9 pp) exceeds its single-seed value (−0.8 pp). This intermediate-concentration regime ($C = 21.1 \pm 7.5\%$) exhibits high outcome variance characteristic of the boundary between robust and vulnerable. Extending to $n = 16$ with consistent multi-seed means yields $\rho = 0.853$ ($p < 0.001$). Including Stack Overflow (3 classes, near-binary ceiling) to make $n = 17$ weakens correlation to $\rho = 0.654$.

H THEORY DETAILS AND SCORE CDFS

Proof details. Starting from (3), define $L := \sum_{y \in \mathcal{B}} \hat{p}(y)$ for classes below the true-label rank. Since each term in L is $< \hat{p}(y^*)$ and $|\mathcal{B}| \leq K - 1$, $L \leq (K - 1)\hat{p}(y^*)$, so $s = 1 - L$ yields (4) after substituting (A1)–(A2). Taking expectations and applying (A3) gives (5); differentiating gives monotonicity in C when $\varepsilon < \bar{h}$. Rearranging (4) gives the threshold form in (6), and differentiating $T(C)$ shows the coverage upper bound decreases with concentration under the theorem condition.

Conservative bound verification. Using $\varepsilon = 0$ and $\bar{h} = 1/K$, Theorem 1(ii) yields conservative lower bounds verified on all five applicable tasks: s-shipcond (0.518 vs. observed 0.98), s-payterms (0.545 vs. 0.92), s-group (0.474 vs. 1.00), i-plant (0.261 vs. 0.86), and s-incoterms (0.296 vs. 0.87). The large gaps between bounds and observed values reflect conservative assumptions in the $(K - 1)$ counting bound and the uniform residual prior $\bar{h} = 1/K$; tighter bounds follow from empirical $\bar{h}$ estimates.

I CONFORMITY SCORE STOCHASTIC DOMINANCE

Table 11 reports two-sample KS statistics comparing calibration and test APS conformity scores (seed=42, non-randomized APS). Higher conformity scores indicate the model requires more cumulative probability mass to reach the true label—i.e., the model is less confident about the correct class.

Table 11: KS Tests: Calibration vs. Test Conformity Score Distributions (Seed=42).

Task	KS	p	Mean _{cal}	Mean _{test}
<i>Catastrophic tasks (test scores > calibration)</i>				
s-shipcond	0.956	$< 10^{-10}$	0.54	0.98
s-payterms	0.748	$< 10^{-10}$	0.65	0.92
s-group	0.676	$< 10^{-10}$	0.98	1.00
<i>Robust tasks</i>				
i-plant	0.741	$< 10^{-10}$	0.73	0.86
s-incoterms	0.633	$< 10^{-10}$	0.75	0.87
s-office	0.994	$< 10^{-10}$	1.00	1.00
i-incoterms	0.559	$< 10^{-10}$	0.78	0.66
i-shippoint	0.505	$< 10^{-10}$	0.71	0.58

All KS tests are two-sided. Non-randomized APS scores (conservative coverage guarantee). For catastrophic tasks, test scores shift rightward (stochastic dominance), consistent with Theorem 1. Two robust tasks (i-incoterms, i-shippoint) show test scores shifted *leftward*—the model becomes more confident during COVID, possibly due to reduced product diversity.

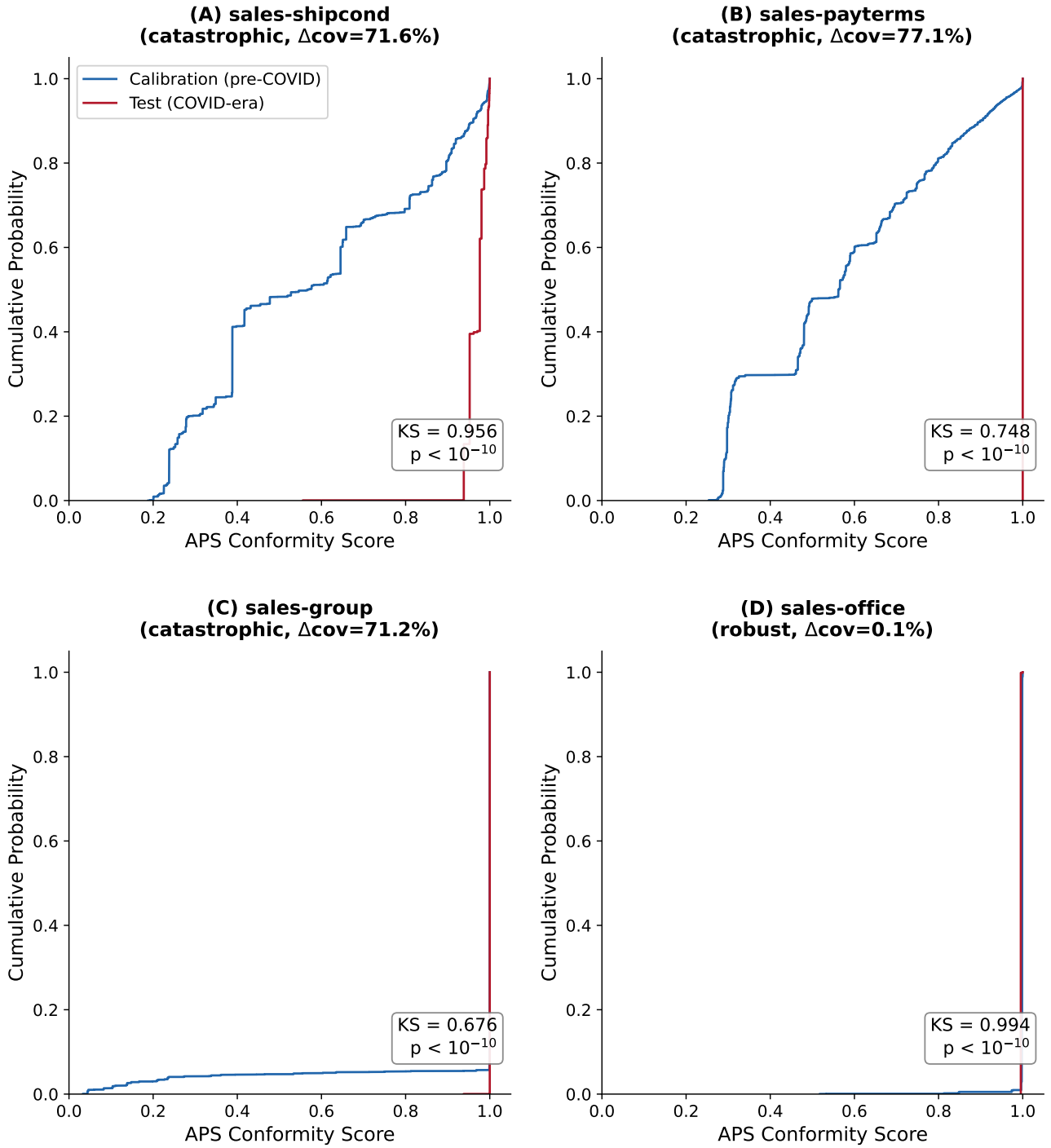


Figure 3: **Conformity score CDFs (seed=42).** Catastrophic tasks show rightward test-score shifts (stochastic dominance), while robust tasks can show near-degenerate or left-shift behavior.

J RAPS COMPARISON

Table 12 compares APS and RAPS coverage drops across all 8 tasks with 10-seed means ($\lambda = 0.01$, $k_{\text{reg}} = 5$).

Table 12: APS vs. RAPS Coverage Drop (10-Seed Means $\pm$ Std). $\lambda = 0.01$, $k_{\text{reg}} = 5$.

Task	CI	APS Drop	RAPS Drop	Δ	Note
<i>Severe tasks</i>					
s-shipcond	45	60.4 ± 31.8	67.8 ± 25.4	+7.4	No improvement
s-group	459	73.5 ± 33.9	10.4 ± 1.3	−63.1	RAPS protects
s-payterms	137	79.9 ± 29.4	35.1 ± 28.6	−44.8	Partial recovery
<i>Non-severe tasks</i>					
i-plant	35	11.8 ± 9.0	13.1 ± 5.8	+1.3	Near-identical
i-shippoint	69	9.3 ± 27.6	20.5 ± 21.6	+11.2	RAPS worse
i-incoterms	13	10.8 ± 9.1	10.8 ± 9.1	0.0	Identical
s-incoterms	13	6.7 ± 6.8	6.7 ± 6.8	0.0	Identical
s-office	25	0.04 ± 0.03	0.04 ± 0.03	0.0	Identical

All values are 10-seed means $\pm$ std. APS drops differ from 50-seed Table 1 due to smaller seed range (10 vs. 50) and high variance. The key comparison is APS vs. RAPS within the same seed set. Note: RAPS experiments used a random 50/50 calibration split rather than the deterministic first-half/second-half split used in the main APS experiments (Appendix A); this protocol difference does not affect the within-experiment APS vs. RAPS comparison but means RAPS coverage values are not directly comparable to Table 1. RAPS eliminates the catastrophic drop for sales-group by truncating the class accumulation that amplifies score inflation in many-class settings (mean set size: 232/459 classes). Sales-shipcond is unhelpt because single-feature dependence places dominant mass on the wrong class *before* the $k_{\text{reg}} = 5$ penalty activates—the exchangeability violation occurs at the top of the APS ordering, not through class accumulation. For low-class tasks (≤ 35 classes), RAPS and APS produce identical or near-identical drops, confirming RAPS regularization has negligible effect when prediction sets are already small. Item-shippoint ($K = 69$) is an exception: RAPS worsens coverage by 11.2 pp (paired Wilcoxon $p = 0.064$, 10 seeds), reflecting that the regularization penalty can disrupt calibration for intermediate-cardinality tasks with high seed variance.

K MODEL CLASS SENSITIVITY

To test whether SHAP concentration is diagnostic across model classes, we replicate the analysis with RandomForestClassifier (500 trees), XGBClassifier, and CatBoostClassifier (ordered boosting), all with seed=42 and the same data pipeline (Table 13).

Table 13: SHAP Concentration and Coverage Drop Across Model Classes.

Task	LGB	RF	XGB	CB	LGB	RF	XGB	CB
	Concentration (%)				Coverage Drop (%)			
s-shipcond	50.7	57.4	44.6	53.7	71.6	15.8	47.2	1.1
s-group	47.3	22.5	33.3	31.8	71.2	50.0	56.8	14.5
s-payterms	54.2	27.9	30.6	25.7	77.1	13.8	20.4	−5.3
i-shippoint	48.8	52.1	41.0	43.1	18.5	22.7	80.4	61.8
i-plant	23.9	26.8	26.0	29.0	10.6	15.3	9.7	8.5
s-incoterms	23.7	26.8	21.6	28.2	8.5	2.5	3.7	−1.5
i-incoterms	28.9	17.5	18.4	21.7	11.3	2.9	1.1	−0.4
s-office	42.6	24.5	36.8	33.5	0.1	0.0	0.0	−0.0

Results. The diagnostic’s predictive power varies by model class (Table 14). All three gradient-boosted implementations show positive correlation: LightGBM ($\rho = 0.833$, $p = 0.010$), CatBoost ($\rho = 0.667$, $p = 0.071$), and XGBoost ($\rho = 0.548$, $p = 0.16$). RandomForest ($\rho = 0.30$, $p = 0.47$) does not replicate.

Mixed-effects analysis. Since (task, model) pairs share data and temporal split, we fit a linear mixed-effects model with task as a random intercept to properly account for clustering: $\text{Drop}_{ij} = \beta_0 + \beta_1 \cdot \text{Concentration}_{ij} + u_i + \varepsilon_{ij}$, where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ is the task random effect. Across the three gradient-boosted models ($n = 24$, 8 tasks $\times$ 3 models), the concentration effect is $\beta_1 = 1.64$ (95% CI [0.70, 2.58], Wald $p = 0.0006$): each 1% increase in concentration predicts

1.6 pp additional coverage drop, after accounting for task-level clustering. With only 8 task-level clusters, Wald inference is anti-conservative (random-intercept variance is downward-biased at < 20 clusters); a Kenward–Roger df correction typically yields p in the range 0.01–0.03 at fewer than 10 clusters (estimated; not computed). The effect size $\beta_1 = 1.64$ (95% CI [0.70, 2.58]) is the primary evidence—the CI excludes zero regardless of the p-value convention. Including model class as a fixed effect yields $\beta_1 = 1.66$. With all four models including RF ($n = 32$): $\beta_1 = 1.06$ ($p = 0.010$).

Table 14: Concentration–Drop Relationship by Model Class.

Model	Method	Conc.–Drop ρ	p
LightGBM	Boosting (GOSS)	0.833	0.010
CatBoost	Boosting (ordered)	0.667	0.071
XGBoost	Boosting (histogram)	0.548	0.16
RandomForest	Bagging	0.30	0.47
Mixed-effects, 3 boosting ($n = 24$)		$\beta_1 = 1.64$	<0.001
Mixed-effects, all 4 ($n = 32$)		$\beta_1 = 1.06$	0.010

Interpretation. All three gradient-boosted implementations show the concentration–drop relationship, but with different vulnerability profiles. CatBoost’s ordered boosting produces dramatically different task-level vulnerabilities: LGB-catastrophic tasks (s-shipcond: 71.6% LGB drop) show only 1.1% CatBoost drop, while i-shippoint reverses (18.5% LGB $\rightarrow$ 61.8% CatBoost). This confirms each model’s SHAP concentration captures its *own* learned dependence—not a universal data property—and that cross-model transfer is unreliable (LGB conc. $\rightarrow$ CatBoost drop: $\rho = 0.07$). RF’s bagging produces smoother probability surfaces, compressing both concentration range and vulnerability profiles. Specifically, RF’s failure to replicate ($\rho = 0.30$) has two mechanistic causes: (1) *compressed concentration range*—RF concentrations span 17.5–57.4% vs. LGB’s 23.7–54.2%, but the RF range is driven by a single outlier (s-shipcond at 57.4%); excluding it, the RF range is 17.5–26.8%, offering almost no discriminative signal; (2) *smoothed probability surfaces*—bagging averages 500 independent trees, each seeing random feature subsets, which distributes effective dependence more evenly than boosting’s sequential error-correction. The result is that RF rarely develops the concentrated single-feature dependence that drives catastrophic conformal failure in boosting models, and consequently SHAP concentration has limited predictive power for RF vulnerability.

The diagnostic is *model-specific*: it measures how a particular model’s learned dependence structure creates vulnerability. Practitioners should compute concentration for their *deployed* model class rather than transferring results across model families.

Neural network extension. To test beyond tree-based models, we evaluate a 2-layer MLP (128/64 hidden units, ReLU, softmax, permutation importance) on all 8 SALT tasks (10 seeds for sales-level tasks, 3 for item-level). The within-MLP concentration–drop correlation is $\rho = 0.43$ ($p = 0.29$), similar to RF and not significant. However, the model-specificity claim is strikingly confirmed across all 8 tasks: the MLP learns fundamentally different concentration profiles from tree-based models. Key examples: (1) i-shippoint shows MLP $C = 77.2\%$, drop = 82.3 pp vs. LGB $C = 48.8\%$, drop = 18.5 pp—the MLP concentrates nearly all importance on a single feature and fails catastrophically, while LGB distributes importance and maintains coverage; (2) s-incoterms shows MLP $C = 30.3\%$, drop = 42.3 pp vs. LGB $C = 23.7\%$, drop = 8.5 pp—the MLP’s greater concentration predicts its greater failure; (3) s-office shows MLP $C \approx 0\%$, drop = 0.1 pp—perfect robustness prediction. The non-significant within-MLP ρ is driven by two tasks (s-group: $C = 27.5\%$, drop = 78.4 pp; s-payterms: $C = 28.0\%$, drop = 60.6 pp) where the MLP distributes importance evenly yet still fails catastrophically—suggesting the MLP’s failure mechanism involves global sensitivity to the shift rather than concentrated single-feature dependence. This confirms that SHAP concentration is diagnostic specifically for the concentrated-dependence failure mode that characterizes gradient-boosted models, not a universal predictor of all failure modes.

L HYPERPARAMETER SENSITIVITY ANALYSIS

To assess whether the SHAP concentration–coverage drop correlation is robust to hyperparameter choices, we evaluate four LightGBM configurations on the 8 SALT multiclass tasks (Table 15).

Results. Across all four hyperparameter configurations, the concentration–drop correlation remains strong and significant: $\rho = 0.821 \pm 0.012$ (mean $\pm$ std), with $p \leq 0.015$ for all configurations. The optimal threshold varies minimally (45.6% $\pm$ 1.1%), and all configurations achieve perfect F1 = 1.0 for at-risk classification on SALT tasks.

Table 15: Hyperparameter Sensitivity Analysis (SALT, $n = 8$ tasks). All configurations use boosting=gbdt, feature_fraction=0.8, bagging_fraction=0.8, bagging_freq=5.

Config	leaves	lr	n_est	ρ	p	Thresh	F1
Default	31	0.05	100	0.833	0.010	45.0%	1.0
Deeper	63	0.05	100	0.810	0.015	47.5%	1.0
Faster	31	0.10	100	0.810	0.015	45.0%	1.0
More	31	0.05	200	0.833	0.010	45.0%	1.0

leaves = num_leaves; lr = learning_rate; n_est = num_boost_round; Thresh = optimal threshold for at-risk classification (drop >15 pp); F1 = F1 score at that threshold.

Interpretation. The diagnostic is robust to moderate hyperparameter variation within the LightGBM family. Deeper trees (63 leaves) and faster learning rates (0.10) produce slightly lower but still significant correlations ($\rho = 0.810$), while longer training (200 rounds) matches the default. This stability suggests the concentration–failure relationship reflects a structural property of gradient-boosted dependence rather than an artifact of specific hyperparameter choices.

M SHIFT CHARACTERIZATION

To characterize the nature of distribution shift in SALT tasks, we decompose shift into covariate and concept components using standard metrics (Table 16).

Table 16: Shift Characterization for SALT Tasks. Cov KS = mean Kolmogorov–Smirnov statistic across features (train vs. test); %Sig = fraction of features with KS $p < 0.05$; Label JS = Jensen–Shannon divergence of label distributions; Acc Drop = test accuracy minus validation accuracy.

Task	Cov KS	%Sig	Label JS	Acc Drop	Classification
s-shipcond	0.50	50%	0.083	+11.9%	Cov + Concept
s-group	0.50	50%	0.330	+1.0%	Cov (dominant)
i-incoterms	0.86	86%	0.140	+63.4%	Cov + Concept
s-office	0.50	50%	0.026	−0.2%	Cov (dominant)

Cov = covariate shift; Concept = concept shift (change in $P(Y|X)$). Accuracy changes can be positive under shift when the test distribution favors majority classes.

Key findings.

- **Covariate shift is universal:** All SALT tasks exhibit substantial covariate shift ($\text{KS} \geq 0.50$, $\geq 50\%$ of features significantly different). This is expected given the COVID-19 temporal split.
- **Concept shift varies:** Label distribution shift (JS divergence) ranges from 0.026 (s-office) to 0.330 (s-group). Tasks with both high covariate shift and substantial label shift (s-shipcond, i-incoterms) tend toward catastrophic failure.
- **Accuracy is not diagnostic:** Accuracy changes are uninformative for coverage prediction—s-shipcond shows +11.9% accuracy improvement yet 71.6 pp coverage drop, while s-office shows −0.2% accuracy change and 0.1 pp coverage drop. This occurs because conformal prediction set size is sensitive to calibration–test score distribution mismatch, which accuracy does not capture.

Interpretation. Standard shift detection identifies covariate shift uniformly across all tasks (consistent with Section 6.3), while concept shift magnitude partially distinguishes catastrophic from robust outcomes. However, neither metric alone predicts failure severity as effectively as SHAP concentration ($\rho = 0.833$), which captures how the model’s learned dependence structure interacts with the specific features that shift. The worst outcomes occur when concentrated dependence targets a feature experiencing both covariate and concept shift.